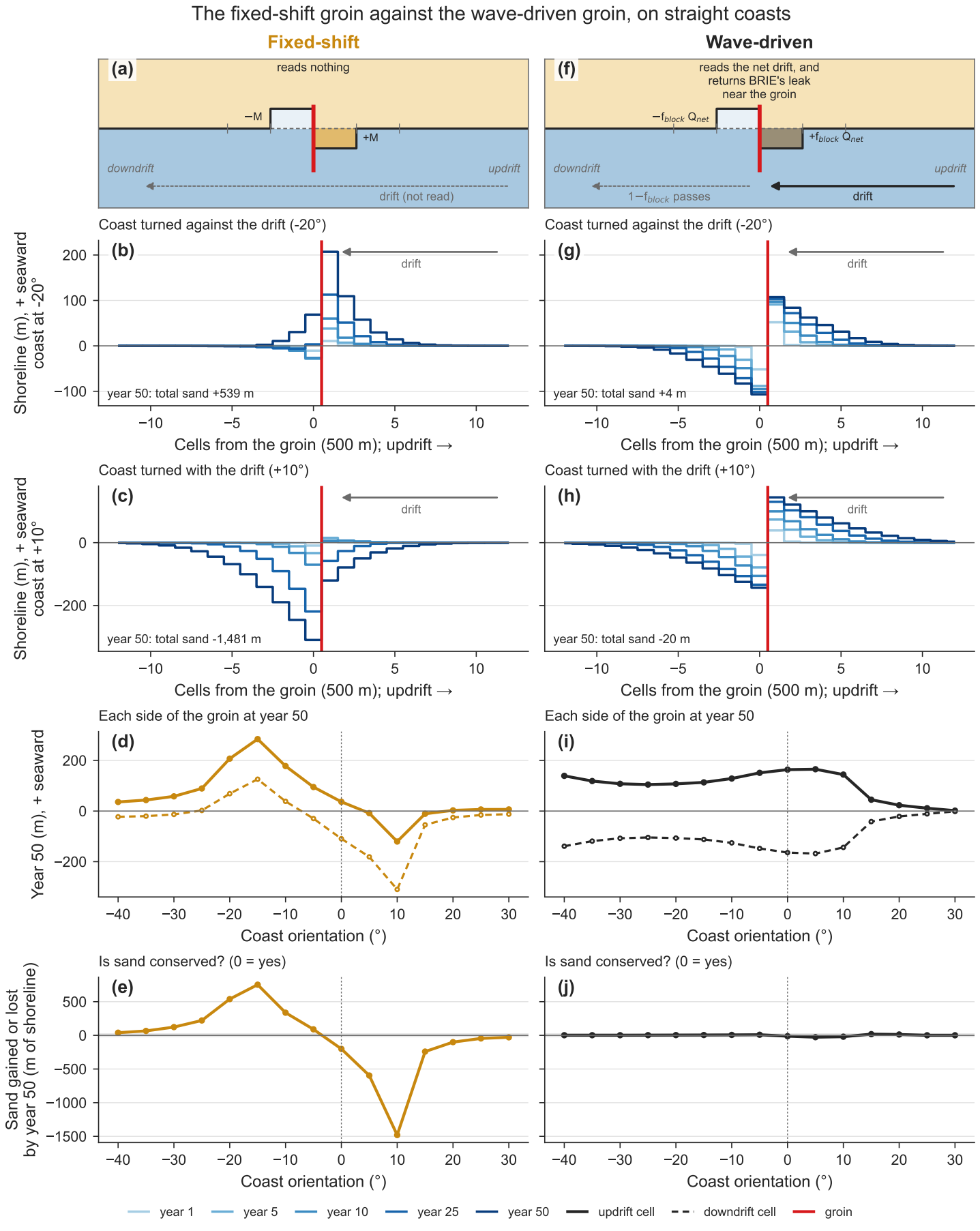
The fixed-shift groin against the wave-driven groin, on straight coasts
Fixed-shift
Wave-driven
(a)
reads nothing
−M
+M
downdrift
updrift
drift (not read)
(f)
reads the net drift, and returns BRIE's leak near the groin
−f_block Q_net
+f_block Q_net
downdrift
updrift
1−f_block passes
drift
Coast turned against the drift (-20°)
(b)
drift
year 50: total sand +539 m
Shoreline (m), + seaward coast at -20°
Cells from the groin (500 m); updrift →
(g)
drift
year 50: total sand +4 m
Cells from the groin (500 m); updrift →
Coast turned with the drift (+10°)
(c)
drift
year 50: total sand -1,481 m
Shoreline (m), + seaward coast at +10°
Cells from the groin (500 m); updrift →
(h)
drift
year 50: total sand -20 m
Cells from the groin (500 m); updrift →
Each side of the groin at year 50
(d)
Year 50 (m), + seaward
Coast orientation (°)
(i)
Coast orientation (°)
Is sand conserved? (0 = yes)
(e)
Sand gained or lost by year 50 (m of shoreline)
Coast orientation (°)
(j)
Coast orientation (°)
year 1
year 5
year 10
year 25
year 50
updrift cell
downdrift cell
groin